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Design Data Pipeline using Databricks(Cloud) and SQL

For this task, I worked on a sales dataset in CSV format to build a data pipeline using SQL and cloud concepts. The main objective was to take raw dataset , clean it and to build a data pipeline using SQL and cloud concepts.

First, I loaded the data into **Databricks using Spark** and stored it in the **Bronze layer** without making any changes. This keeps the original data.

Next, in the **Silver layer**, I cleaned the data by **removing null values, removing duplicates, and fixing the format**. This made the data ready for analysis.

Then, in the **Gold layer**, I used SQL to create summary tables and calculate total sales and other values.

Finally, I used Databricks SQL to analyse the data and create simple charts to see sales trends.

Then , created an **architecture diagram using draw.io to show the data flow**. The diagram includes the data source (CSV file), cloud storage (Data Lake), Databricks as the compute layer, and Bronze, Silver, and Gold layers.

This pipeline follows Bronze, Silver, and Gold layers and uses cloud storage and distributed.